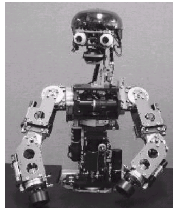
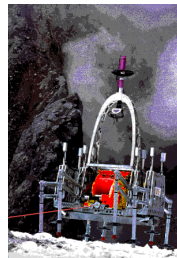


The 3D kinematic problem in different robots

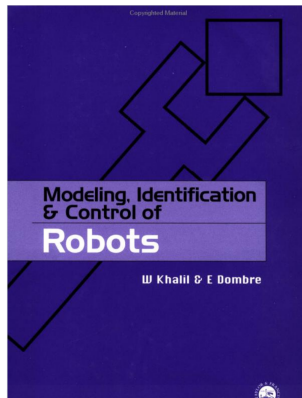
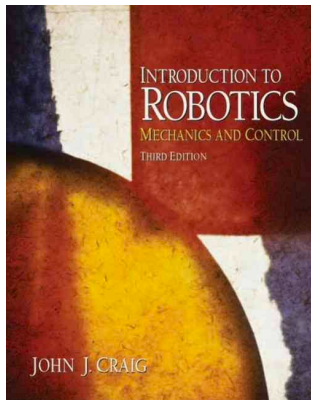


The ability to know the location and shape of limbs relative to a common base is crucial for meaningful behavior of all these robots.

3D kinematics - learning outcomes

- Know the structure and properties of a 3D rotation matrix
- Know what a homogeneous coordinate system is
- Know the structure of a homogeneous transformation matrix
- Know what is meant by linearly independent axes are
- Know the physical meaning of an Euler rotation
- Know the difference between successive Euler rotations around moving axes (Euler angles) and fixed axes (Roll, Yaw, and Pitch)
- Know what a unit quaternion is and how to perform 3D rotations with quaternions
- Know the properties of homogeneous transformation matrices and their products
- Know how to perform a compound transformation (a series of 3D transformations)
- Know how to obtain the Jacobian matrix of a manipulator from the compound homogeneous transformation matrix

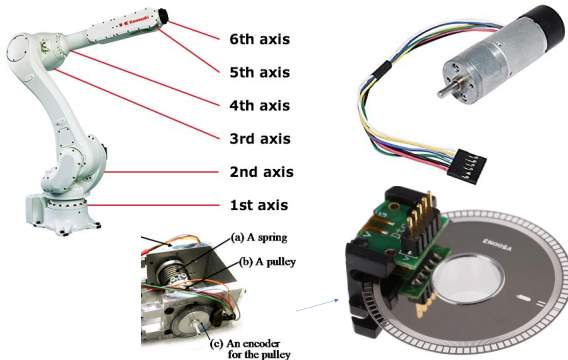
Relevant books



Please note that the given booklet is enough for this module. There, I have given all essential information in a compact form.

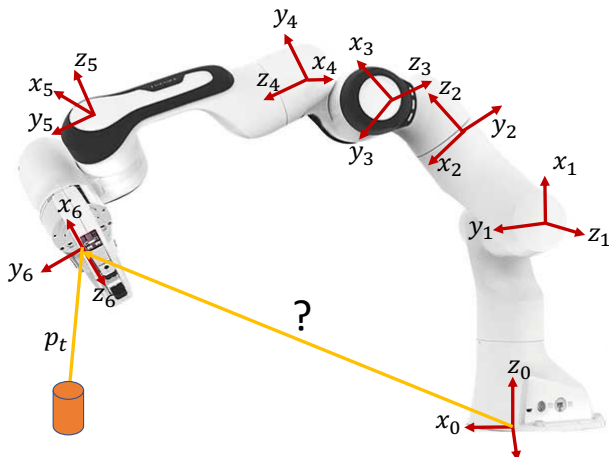
Joints and joint encoders

The important thing to remember is that each joint only knows how much the link attached to it rotates based on the joint encoders at each joint.

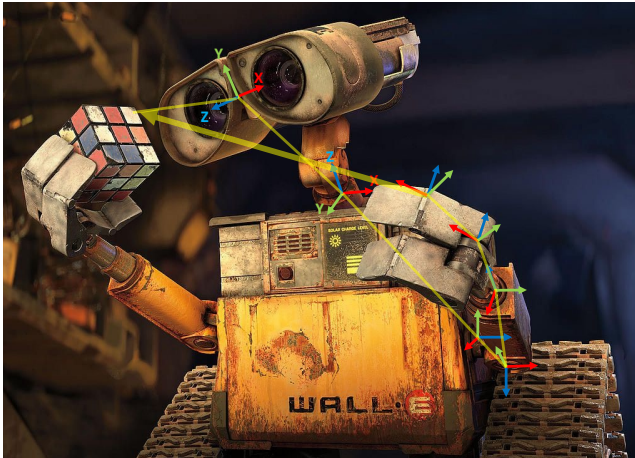


The forward kinematics problem

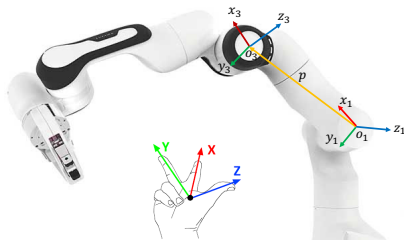
Given knowledge of only the joint angles and link lengths, how to determine the position and orientation of the end-effector of a robot relative to the base.



The forward kinematics problem



Frames



A coordinate frame i consists of an origin, denoted O_i , and three mutually orthogonal basis vectors, denoted (x_i, y_i, z_i) , that are fixed within a given body.

Homogeneous coordinate system

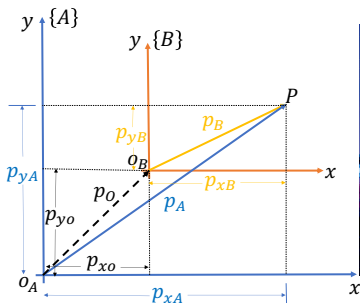
$$\mathbf{p}_1 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (1)$$

Vector $\mathbf{p}_1$ is in Cartesian coordinates because the origin is fixed at $(0, 0, 0)$.

$$\mathbf{p}_2 = \begin{bmatrix} kx \\ ky \\ kz \\ k \end{bmatrix} \quad (2)$$

Vector $\mathbf{p}_2$ is in homogeneous coordinates because the origin depends on k .

Pure translation in homogeneous coordinates



$$\begin{bmatrix} p_{xA} \\ p_{yA} \\ 1 \end{bmatrix} = \begin{bmatrix} p_{x0} \\ p_{y0} \\ 1 \end{bmatrix} + \begin{bmatrix} p_{xB} \\ p_{yB} \\ 1 \end{bmatrix}$$

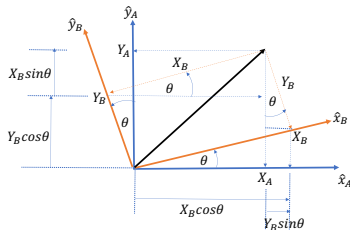
$$= \begin{bmatrix} p_{x0} + p_{xB} \\ p_{y0} + p_{yB} \\ 1 \end{bmatrix}$$

(3)

Pure translation in homogeneous coordinates

$$\begin{aligned}\begin{bmatrix} p_{xA} \\ p_{yA} \\ 1 \end{bmatrix} &= \begin{bmatrix} p_{x0} + p_{xB} \\ p_{y0} + p_{yB} \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times p_{xB} + 0 \times p_{yB} + 1 \times p_{x0} \\ 0 \times p_{xB} + 1 \times p_{yB} + 1 \times p_{y0} \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & p_{x0} \\ 0 & 1 & p_{y0} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{xB} \\ p_{yB} \\ 1 \end{bmatrix} \quad (4)\end{aligned}$$

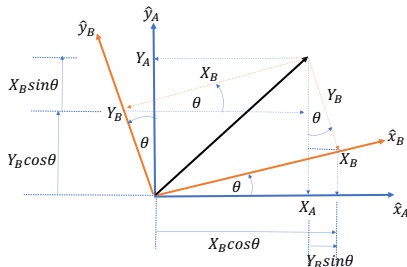
Geometric interpretation of rotation matrix in 2D



$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} X_B \cos \theta - Y_B \sin \theta \\ X_B \sin \theta + Y_B \cos \theta \\ 1 \end{bmatrix} \quad (5)$$

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (6)$$

Vector dot product (projection) interpretation of rotation matrix in 2D

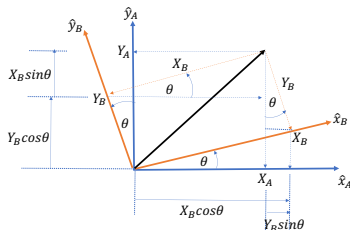


It can also be obtained by dot products of unit vectors along each axis:

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \hat{x}_B \cdot \hat{x}_A & \hat{y}_B \cdot \hat{x}_A & 0 \\ \hat{x}_B \cdot \hat{y}_A & \hat{y}_B \cdot \hat{y}_A & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (7)$$

where, $\hat{x}_A$, $\hat{x}_B$, $\hat{y}_A$, and $\hat{y}_B$ are unit vectors (magnitude = 1).

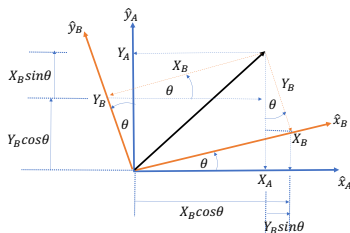
Vector dot product (projection) interpretation of rotation matrix in 2D



$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \hat{x}_B \cdot \hat{x}_A & \hat{y}_B \cdot \hat{x}_A & 0 \\ \hat{x}_B \cdot \hat{y}_A & \hat{y}_B \cdot \hat{y}_A & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (8)$$

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \cos(90 + \theta) & 0 \\ \cos(90 - \theta) & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (9)$$

Vector dot product (projection) interpretation of rotation matrix in 2D

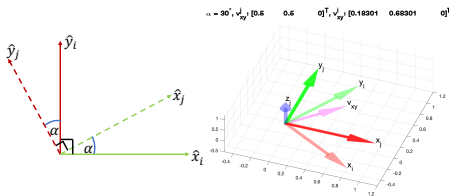


$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \cos(90 + \theta) & 0 \\ \cos(90 - \theta) & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (10)$$

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix} \quad (11)$$

Rotation around $\hat{z}$ axis

$\hat{z}$ axis is coming out of the paper. Check z-axis section of [ZYXindividualRotations.m](#)

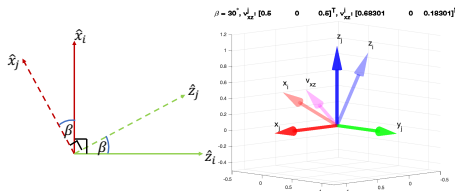


$${}^i R_j(\alpha) = \begin{bmatrix} |\hat{x}_j||\hat{x}_i| \cos(\alpha) & |\hat{y}_j||\hat{x}_i| \cos(90 + \alpha) & |\hat{z}_j||\hat{x}_i| \cos(90) \\ |\hat{x}_j||\hat{y}_i| \cos(90 - \alpha) & |\hat{y}_j||\hat{y}_i| \cos(\alpha) & |\hat{z}_j||\hat{y}_i| \cos(90) \\ |\hat{x}_j||\hat{z}_i| \cos(90) & |\hat{y}_j||\hat{z}_i| \cos(90) & |\hat{z}_j||\hat{z}_i| \cos(0) \end{bmatrix} \quad (12)$$

$$R_{\hat{z}}(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (13)$$

Rotation around $\hat{y}$ axis

$\hat{y}$ axis is coming out of the paper. Check y-axis section of [ZYXindividualRotations.m](#)

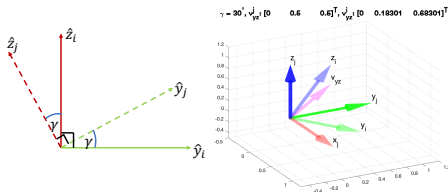


$${}^i\mathbf{R}_j(\beta) = \begin{bmatrix} |\hat{\mathbf{x}}_j||\hat{\mathbf{x}}_i| \cos(\beta) & |\hat{\mathbf{y}}_j||\hat{\mathbf{x}}_i| \cos(90) & |\hat{\mathbf{z}}_j||\hat{\mathbf{x}}_i| \cos(90 - \beta) \\ |\hat{\mathbf{x}}_j||\hat{\mathbf{y}}_i| \cos(90) & |\hat{\mathbf{y}}_j||\hat{\mathbf{y}}_i| \cos(0) & |\hat{\mathbf{z}}_j||\hat{\mathbf{y}}_i| \cos(90) \\ |\hat{\mathbf{x}}_j||\hat{\mathbf{z}}_i| \cos(90 + \beta) & |\hat{\mathbf{y}}_j||\hat{\mathbf{z}}_i| \cos(90) & |\hat{\mathbf{z}}_j||\hat{\mathbf{z}}_i| \cos(\beta) \end{bmatrix} \quad (14)$$

$$\mathbf{R}_{\hat{y}}(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \quad (15)$$

Rotation around $\hat{x}$ axis

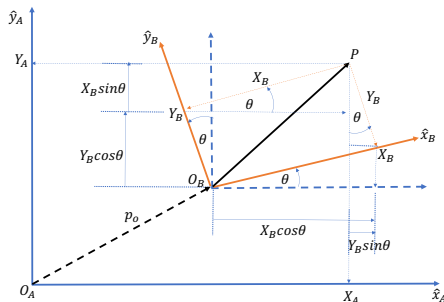
$\hat{x}$ axis is coming out of the paper. Check x-axis section of [ZYXindividualRotations.m](#)



$${}^iR_j(\gamma) = \begin{bmatrix} |\hat{x}_j||\hat{x}_i| \cos(0) & |\hat{y}_j||\hat{x}_i| \cos(90) & |\hat{z}_j||\hat{x}_i| \cos(90) \\ |\hat{x}_j||\hat{y}_i| \cos(90) & |\hat{y}_j||\hat{y}_i| \cos(\gamma) & |\hat{z}_j||\hat{y}_i| \cos(90 + \gamma) \\ |\hat{x}_j||\hat{z}_i| \cos(90) & |\hat{y}_j||\hat{z}_i| \cos(90 - \gamma) & |\hat{z}_j||\hat{z}_i| \cos(\gamma) \end{bmatrix} \quad (16)$$

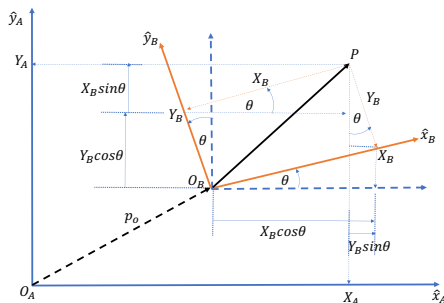
$$R_{\hat{x}}(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix} \quad (17)$$

2D rotation and translation



$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} & \begin{bmatrix} p_{x0} \\ p_{y0} \\ 1 \end{bmatrix} \end{bmatrix} \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix}$$

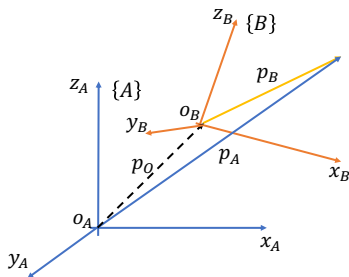
2D rotation and translation



Which is the same as

$$\begin{bmatrix} X_A \\ Y_A \\ 1 \end{bmatrix} = \begin{bmatrix} \hat{x}_B \cdot \hat{x}_A & \hat{y}_B \cdot \hat{x}_A \\ \hat{x}_B \cdot \hat{y}_A & \hat{y}_B \cdot \hat{y}_A \\ 0 & 0 \end{bmatrix} \begin{bmatrix} p_{x0} \\ p_{y0} \\ 1 \end{bmatrix} + \begin{bmatrix} X_B \\ Y_B \\ 1 \end{bmatrix}$$

Rotation and translation in 3D



$${}^A\mathbf{p} = \begin{bmatrix} p_{xA} \\ p_{yA} \\ p_{zA} \end{bmatrix}, \quad \mathbf{p}_o = \begin{bmatrix} p_{xO} \\ p_{yO} \\ p_{zO} \end{bmatrix}, \quad {}^B\mathbf{p} = \begin{bmatrix} p_{xB} \\ p_{yB} \\ p_{zB} \end{bmatrix} \quad (18)$$

Rotation and translation in 3D

$$\begin{bmatrix} p_{xA} \\ p_{yA} \\ p_{zA} \\ 1 \end{bmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & p_{xO} \\ R_{21} & R_{22} & R_{23} & p_{yO} \\ R_{31} & R_{32} & R_{33} & p_{zO} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_{xB} \\ p_{yB} \\ p_{zB} \\ 1 \end{bmatrix} \quad (19)$$

$$\begin{bmatrix} {}^A\mathbf{p} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{p}_B \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^B\mathbf{p} \\ 1 \end{bmatrix} \quad (20)$$

Transformation matrix:

$${}^A\mathbf{T}_B = \begin{bmatrix} R_{11} & R_{12} & R_{13} & p_{xO} \\ R_{21} & R_{22} & R_{23} & p_{yO} \\ R_{31} & R_{32} & R_{33} & p_{zO} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} {}^A\mathbf{R}_B & {}^A\mathbf{p}_B \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \quad (21)$$

Rotation matrix from frame $(\hat{x}_j, \hat{y}_j, \hat{z}_j)$ to frame $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ is given by

$${}^iR_j = \begin{bmatrix} \hat{x}_j \cdot \hat{x}_i & \hat{y}_j \cdot \hat{x}_i & \hat{z}_j \cdot \hat{x}_i \\ \hat{x}_j \cdot \hat{y}_i & \hat{y}_j \cdot \hat{y}_i & \hat{z}_j \cdot \hat{y}_i \\ \hat{x}_j \cdot \hat{z}_i & \hat{y}_j \cdot \hat{z}_i & \hat{z}_j \cdot \hat{z}_i \end{bmatrix} \quad (22)$$

where $(\hat{x}_i, \hat{y}_i, \hat{z}_i)$ are the basis vectors (unit vectors along each axis are basis vectors) of frame i , and $(\hat{x}_j, \hat{y}_j, \hat{z}_j)$ are those of frame j .

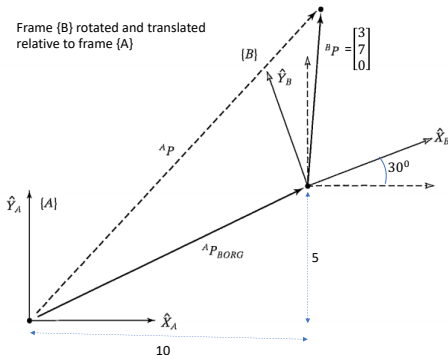
Properties of a rotation matrix

- 1 The **determinant** of a rotation matrix iR_j is +1, because the size of a vector does not change under rotation. i.e. When $v_i = {}^iR_j v_j$, $|v_i| = |v_j|$.
- 2 A rotation matrix is orthonormal. This means, when iR_j is written as three column vectors ${}^iR_j = [R_1 \ R_2 \ R_3]$, $|R_1| = 1$, $|R_2| = 1$, $|R_3| = 1$, and $R_1 \cdot R_2 = 0$, $R_1 \cdot R_3 = 0$, and $R_2 \cdot R_3 = 0$.
- 3 The inverse of a rotation matrix iR_j is equal to its transpose given by ${}^iR_j^T$. i.e. Given $v_i = {}^iR_j v_j$, $v_j = {}^iR_j^T v_i$. This also means, ${}^iR_j {}^iR_j^T = I$.
- 4 Multiplication of several rotation matrices results in a rotation matrix. Therefore, rotations can be broken down into intermediate steps when required such as ${}^A R_D = {}^A R_B {}^B R_C {}^C R_D$.

Run the second section of `zyxEuler.m` to check these properties.

Exercise-1

In a general case: frame $\{B\}$ is rotated relative to frame $\{A\}$ around axis $\hat{z}$ by 30° , and translated by 10 units along $\hat{x}$, 5 units along $\hat{y}$. Find the position of point $P = [3 \ 7 \ 0]$ in frame $\{B\}$ relative to frame $\{A\}$. (Hint: Refer to equation 20)



Solution

In a general case: frame $\{B\}$ is rotated relative to frame $\{A\}$ around axis $\hat{z}$ by 30° , and translated by 10 units along $\hat{x}$, 5 units along $\hat{y}$. Find the position of point $P = [3 \ 7 \ 0]$ in frame $\{B\}$ relative to frame $\{A\}$.

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_B \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} \quad (23)$$

$${}^A T_B = \begin{bmatrix} {}^A R_B & {}^A P_B \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} \cos(30) & -\sin(30) & 0 & 10 \\ \sin(30) & \cos(30) & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (24)$$

Extend [example3Dtransformation.m](#) to obtain the answer.

Euler angles

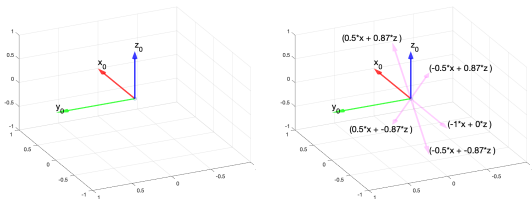
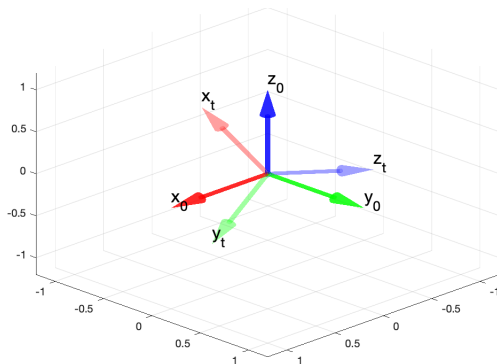


Figure: Idea of linear combinations of vectors. Run [LinearlyDependent.m](#)

“Any 3D rotation can be described by **three successive rotations** around **linearly independent axes**.”

When two vectors v_1 and v_2 are linearly combined in the form $v = c_1 v_1 + c_2 v_2$, where c_1 and c_2 are scalar coefficients, v will be in the same plane of v_1 and v_2 . Any vector outside this plane will be linearly independent of v_1 and v_2 .

ZYX moving frame rotation

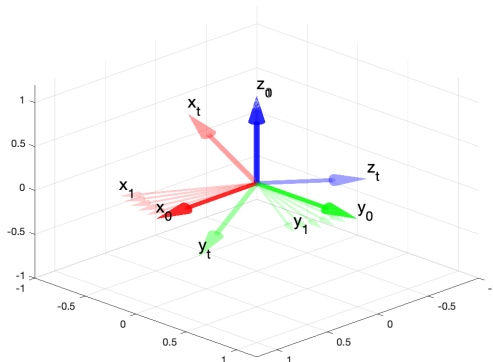


$${}^0R_t = {}^0R_1(\alpha) {}^1R_2(\beta) {}^2R_t(\gamma) \quad (25)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

ZYX moving frame rotation

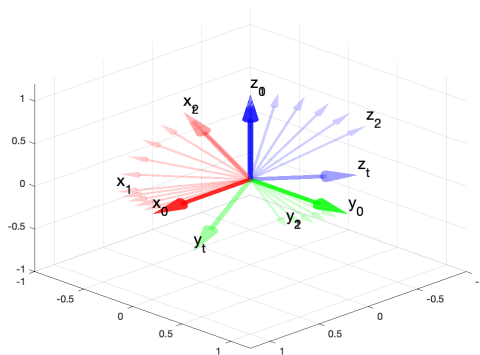
Run [visualizeZYXEuler.m](#).



$${}^0R_t = {}^0R_1(\alpha) {}^1R_2(\beta) {}^2R_t(\gamma) \quad (26)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

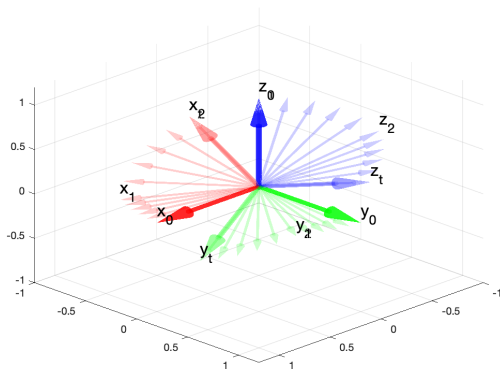
ZYX moving frame rotation



$${}^0R_t = {}^0R_1(\alpha) {}^1R_2(\beta) {}^2R_t(\gamma) \quad (27)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

ZYX moving frame rotation



$${}^0R_t = {}^0R_1(\alpha) {}^1R_2(\beta) {}^2R_t(\gamma) \quad (28)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

The objective of the first two rotations is to get one axis to coincide with the corresponding axis of the target frame. Then, since the three axes are orthogonal, one final rotation around the axis that coincides with the target axis will make all three axes coincide.

In $Z - Y - X$ Euler rotations, frame 1 is rotated by α around $\hat{z}_0$ to get $\hat{x}_1$ and $\hat{x}_t$ to be on the plane perpendicular to $\hat{y}_1$.

Then, a rotation around the new $\hat{y}_1$ by an angle β will get the two x-axes to coincide.

Since this automatically makes the $Y - Z$ planes coincide, a final γ rotation around the new $\hat{x}_2$ axis will finally get all three axes to coincide with frame t .

[Run visualizeZYXEuler.m](#)

Z-Y-X Euler angles

Given a rotation matrix (run `zyxEuler.m`), can we compute the Euler angles?

$$\begin{aligned} {}^0\mathbf{R}_t(\alpha, \beta, \gamma) &= \begin{bmatrix} c(\alpha) c(\beta) & -c(\gamma) s(\alpha) + c(\alpha) s(\beta) s(\gamma) & s(\alpha) s(\gamma) + c(\alpha) c(\gamma) s(\beta) \\ c(\beta) s(\alpha) & c(\alpha) c(\gamma) + s(\alpha) s(\beta) s(\gamma) & -c(\alpha) s(\gamma) + c(\gamma) s(\alpha) s(\beta) \\ -s(\beta) & c(\beta) s(\gamma) & c(\beta) c(\gamma) \end{bmatrix} \\ &= \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \end{aligned} \quad (29)$$

We can use $R_{31} = -s(\beta)$ to compute $\beta = -\arcsin(R_{31})$. Then we can use $R_{11} = c(\alpha)c(\beta)$ and $R_{21} = c(\beta)s(\alpha)$ to compute α using

$$\alpha = \operatorname{atan2}\left(\frac{R_{21}}{c(\beta)}, \frac{R_{11}}{c(\beta)}\right).$$

Similarly we can compute

$$\gamma = \operatorname{atan2}\left(\frac{R_{32}}{c(\beta)}, \frac{R_{33}}{c(\beta)}\right).$$

Knowing

$${}^0\mathbf{R}_t = \begin{bmatrix} \hat{\mathbf{x}}_t \cdot \hat{\mathbf{x}}_0 & \hat{\mathbf{y}}_t \cdot \hat{\mathbf{x}}_0 & \hat{\mathbf{z}}_t \cdot \hat{\mathbf{x}}_0 \\ \hat{\mathbf{x}}_t \cdot \hat{\mathbf{y}}_0 & \hat{\mathbf{y}}_t \cdot \hat{\mathbf{y}}_0 & \hat{\mathbf{z}}_t \cdot \hat{\mathbf{y}}_0 \\ \hat{\mathbf{x}}_t \cdot \hat{\mathbf{z}}_0 & \hat{\mathbf{y}}_t \cdot \hat{\mathbf{z}}_0 & \hat{\mathbf{z}}_t \cdot \hat{\mathbf{z}}_0 \end{bmatrix} \quad (30)$$

and

$${}^0\mathbf{R}_t(\alpha, \beta, \gamma) = \begin{bmatrix} c(\alpha) c(\beta) & -c(\gamma) s(\alpha) + c(\alpha) s(\beta) s(\gamma) & s(\alpha) s(\gamma) + c(\alpha) c(\gamma) s(\beta) \\ c(\beta) s(\alpha) & c(\alpha) c(\gamma) + s(\alpha) s(\beta) s(\gamma) & -c(\alpha) s(\gamma) + c(\gamma) s(\alpha) s(\beta) \\ -s(\beta) & c(\beta) s(\gamma) & c(\beta) c(\gamma) \end{bmatrix}$$

can you compute the angle between the z_0 and z_t axes?

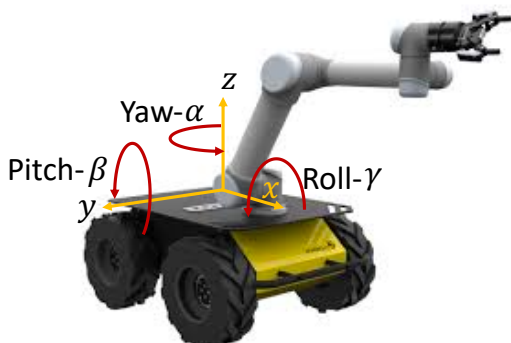
Answer: Let angle between the z_0 and z_t axes be ψ . Then

$$\psi = \arccos(c(\beta)c(\gamma)).$$

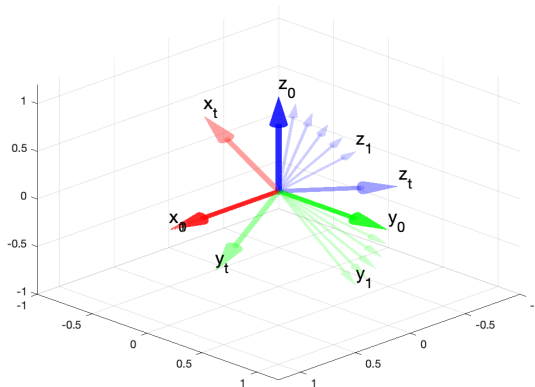
X-Y-Z Fixed angle rotations: Roll, Yaw, and Pitch

Roll, Yaw, and Pitch are often used to denote orientation of robotic links.

- Roll: Rotation around the x-axis (γ)
- Yaw: Rotation around the z-axis (α)
- Pitch: Rotation around the y-axis (β)



X-Y-Z Fixed angle rotations

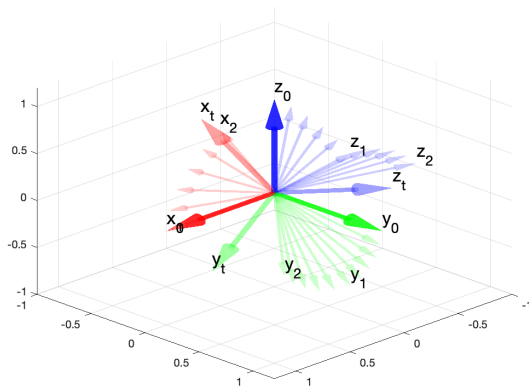


$${}^0R_t(\gamma, \beta, \alpha) = R_{z_0}(\alpha) R_{y_0}(\beta) \mathbf{R}_{x_0}(\gamma) \quad (31)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

Run visualizeXYZfixed.m

X-Y-Z Fixed angle rotations

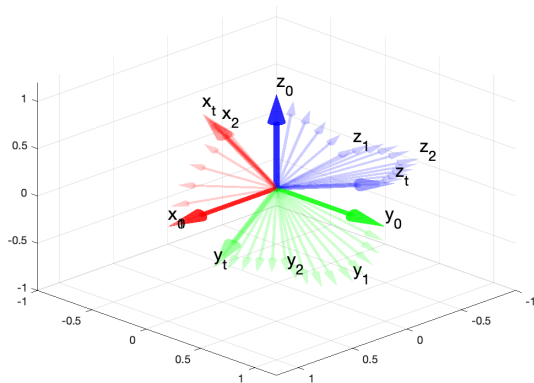


$${}^0R_t(\gamma, \beta, \alpha) = R_{z_0}(\alpha) \mathbf{R}_{y_0}(\beta) R_{x_0}(\gamma) \quad (32)$$

$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

Run visualizeXYZfixed.m

X-Y-Z Fixed angle rotations



$${}^0R_t(\gamma, \beta, \alpha) = \mathbf{R}_{z_0}(\alpha) \mathbf{R}_{y_0}(\beta) \mathbf{R}_{x_0}(\gamma) \quad (33)$$

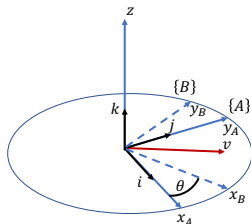
$${}^0R_t = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

Run visualizeXYZfixed.m

Euler rotation around a unit axis

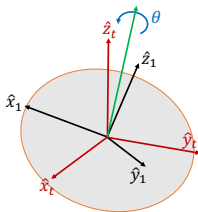
A vector ${}^B\mathbf{v}$ in frame $\{B\}$ can be rotated to ${}^A\mathbf{v}$ in frame $\{A\}$ by a general Euler rotation given by

${}^A\mathbf{v} = \mathbf{q} {}^B\mathbf{v} \mathbf{q}^*$ where $\mathbf{q} = e^{\frac{\theta}{2}\mathbf{u}}$ and $\mathbf{q}^* = e^{-\frac{\theta}{2}\mathbf{u}}$, where $\mathbf{u}$ is a unit vector around which the rotation happens.



For instance, a rotation of a 2D vector in $X - Y$ plane around the z axis is given by $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j} + \mathbf{k}$.

Rotation with quaternions



The idea behind quaternions is that we can find a unit vector

$$\mathbf{u} = k_1 i + k_2 j + k_3 k$$

called the Euler axis in the base frame such that a rotation by an angle θ around $\mathbf{u}$ achieves a 3D rotation. Note that $\hat{K} = [k_1, k_2, k_3]^T$ is a unit vector.

Quaternion

This can be written in the form of an Euler rotation of a vector $\mathbf{p}$ in the base frame to $\mathbf{p}'$ given by $\mathbf{p}' = \mathbf{q} \mathbf{p} \mathbf{q}^*$ where $\mathbf{q} = e^{\frac{\theta}{2}\mathbf{u}}$ and $\mathbf{q}^* = e^{-\frac{\theta}{2}\mathbf{u}}$. $\mathbf{q}^*$ is called the conjugate of $\mathbf{q}$.

$$\begin{aligned}\mathbf{q} &= e^{\frac{\theta}{2}\mathbf{u}} \\ &= e^{\frac{\theta}{2}(k_1i + k_2j + k_3k)} \\ &= \cos \frac{\theta}{2} + (k_1i + k_2j + k_3k) \sin \frac{\theta}{2}\end{aligned}\tag{34}$$

(35)

We define four parameters called Euler parameters: $\epsilon_1 = k_1 \sin \frac{\theta}{2}$, $\epsilon_2 = k_2 \sin \frac{\theta}{2}$, $\epsilon_3 = k_3 \sin \frac{\theta}{2}$, and $\epsilon_4 = \cos \frac{\theta}{2}$.

Quaternion

Then, vector q can be written as

$$q = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4$$

This gives a constraint

$$\begin{aligned}\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 &= (k_1^2 + k_2^2 + k_3^2) \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} \\ &= 1\end{aligned}\tag{36}$$

because $k_1^2 + k_2^2 + k_3^2 = 1$ for the unit vector $\hat{K}$. Since $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$, the 4 element vector $[\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4]$ is called the **unit quaternion**.

Rotation with unit quaternion

Then a 3D rotation around $\mathbf{q}$ vector is given by

$$\begin{aligned} {}^A\mathbf{v} &= \mathbf{R}_\epsilon(\theta) {}^B\mathbf{v} \\ &= \mathbf{q} {}^B\mathbf{v} \mathbf{q}^* \end{aligned} \quad (37)$$

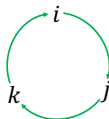
where,

$$\mathbf{q} = \epsilon_1 i + \epsilon_2 j + \epsilon_3 k + \epsilon_4 \quad (38)$$

$$\mathbf{q}^* = -\epsilon_1 i - \epsilon_2 j - \epsilon_3 k + \epsilon_4 \quad (39)$$

$${}^B\mathbf{v} = {}^Bv_x i + {}^Bv_y j + {}^Bv_z k \quad (40)$$

Please note $i*i = j*j = k*k = -1$,
 $i*j = k$, $j*k = i$, $k*i = j$, and
 $j*i = -k$, $k*j = -i$, $i*k = -j$.



Rotation with unit quaternion

Then the rotation matrix is given by

$$\mathbf{R}_\epsilon(\theta) = \begin{bmatrix} 1 - 2\epsilon_2^2 - 2\epsilon_3^2 & 2(\epsilon_1\epsilon_2 - \epsilon_3\epsilon_4) & 2(\epsilon_1\epsilon_3 + \epsilon_2\epsilon_4) \\ 2(\epsilon_1\epsilon_2 + \epsilon_3\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_3^2 & 2(\epsilon_2\epsilon_3 - \epsilon_1\epsilon_4) \\ 2(\epsilon_1\epsilon_3 - \epsilon_2\epsilon_4) & 2(\epsilon_2\epsilon_3 + \epsilon_1\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_2^2 \end{bmatrix}$$

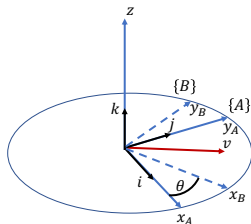
Given a rotation matrix, the equivalent Euler parameters are given by

$$\epsilon_1 = \frac{R_{32} - R_{23}}{4\epsilon_4}, \epsilon_2 = \frac{R_{13} - R_{31}}{4\epsilon_4}, \epsilon_3 = \frac{R_{21} - R_{12}}{4\epsilon_4}, \text{ and}$$

$$\epsilon_4 = \frac{1}{2} \sqrt{1 + R_{11} + R_{22} + R_{33}}.$$

Run Quaternion.m

2D rotation with quaternions



$$\begin{aligned} \mathbf{q} &= \epsilon_4 + k\epsilon_3 \\ &= \cos \frac{\theta}{2} + k \sin \frac{\theta}{2} \end{aligned} \quad (41)$$

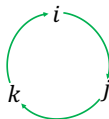
because $k_1 = 0$, $k_2 = 0$, and $k_3 = 1$ to satisfy $k_1^2 + k_2^2 + k_3^2 = 1$. Then, given

$${}^B v = {}^B v_x i + {}^B v_y j$$

2D Euler rotation around a unit axis

$$\begin{aligned} {}^A\mathbf{v} &= \mathbf{q} {}^B\mathbf{v} \mathbf{q}^* \\ &= \left(\cos \frac{\theta}{2} + k \sin \frac{\theta}{2} \right) ({}^Bv_x i + {}^Bv_y j + 0 k) \left(\cos \frac{\theta}{2} - k \sin \frac{\theta}{2} \right) \end{aligned} \quad (42)$$

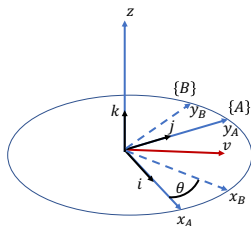
Please note $i*i = j*j = k*k = -1$,
 $i*j = k$, $j*k = i$, $k*i = j$, and
 $j*i = -k$, $k*j = -i$, $i*k = -j$.



By expanding, we obtain

$$\begin{aligned} A_v &= \left(\cos \frac{\theta}{2} + k \sin \frac{\theta}{2} \right) ({}^B v_x i + {}^B v_y j) \left(\cos \frac{\theta}{2} - k \sin \frac{\theta}{2} \right) \\ &= i \left({}^B v_x \left(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) - 2 {}^B v_y \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) \\ &\quad + j \left({}^B v_y \left(\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) + 2 {}^B v_x \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right) \\ &= i ({}^B v_x \cos \theta - {}^B v_y \sin \theta) + j ({}^B v_x \sin \theta + {}^B v_y \cos \theta) \quad (43) \end{aligned}$$

2D Euler rotation around a unit axis



We can then re-arrange the terms to form the rotation matrix given by

$${}^A v = i ({}^B v_x \cos \theta - {}^B v_y \sin \theta) + j ({}^B v_x \sin \theta + {}^B v_y \cos \theta) \quad (44)$$

$$\begin{bmatrix} {}^A v_x \\ {}^A v_y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} {}^B v_x \\ {}^B v_y \end{bmatrix} \quad (45)$$

Properties of transformation matrices

Property-1: The inverse of a transformation matrix is also a transformation matrix and has the following form.

$$T^{-1} = \begin{bmatrix} \mathbf{R} & \mathbf{P} \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^T & -\mathbf{R}^T \mathbf{P} \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \quad (46)$$

Property-2: The product of two transformation matrices is also a transformation matrix.

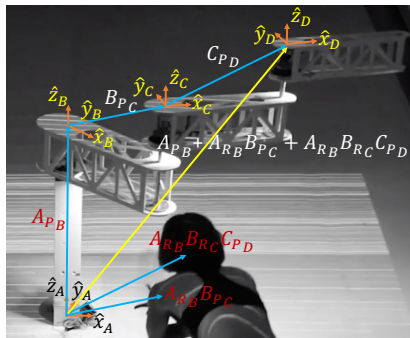
$$T_1 T_2 = \begin{bmatrix} \mathbf{R}_1 & \mathbf{P}_1 \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_2 & \mathbf{P}_2 \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_1 \mathbf{R}_2 & \mathbf{R}_1 \mathbf{P}_2 + \mathbf{P}_1 \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \quad (47)$$

Property-3: Multiplication of transformation matrices is associative, $(T_1 T_2) T_3 = T_1 (T_2 T_3)$, but generally not commutative, $T_1 T_2 \neq T_2 T_1$

Following these properties, compound transformations

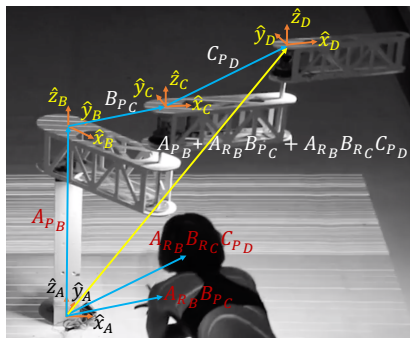
$${}^A\mathbf{T}_D = ({}^A\mathbf{T}_B {}^B\mathbf{T}_C) {}^C\mathbf{T}_D = {}^A\mathbf{T}_B ({}^B\mathbf{T}_C {}^C\mathbf{T}_D)$$

$${}^i\mathbf{T}_j = \begin{bmatrix} {}^i\mathbf{R}_j & {}^i\mathbf{P}_j \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (48)$$



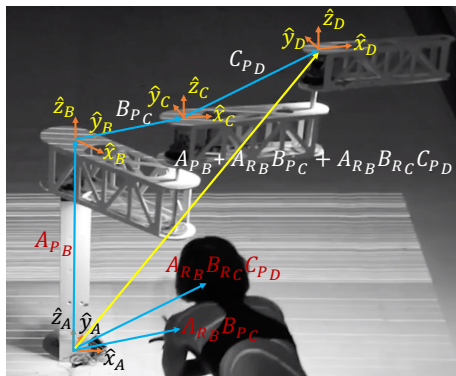
Compound transformations

Example: ${}^B P_C$ is read as the vector to the origin of frame C in frame B .



$$\begin{aligned}
 {}^A T_D &= {}^A T_B {}^B T_C {}^C T_D \\
 &= \begin{bmatrix} {}^A R_B & {}^A P_B \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^B R_C & {}^B P_C \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} {}^C R_D & {}^C P_D \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix}
 \end{aligned} \tag{49}$$

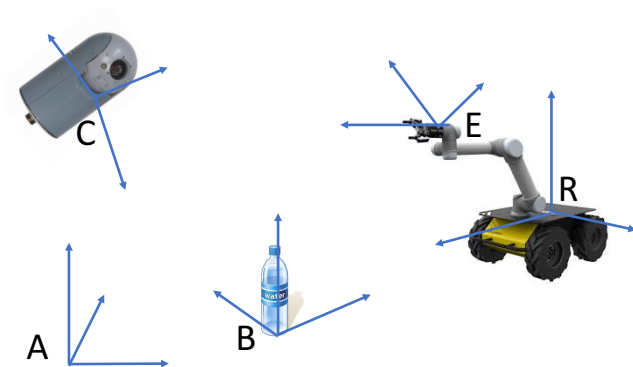
Compound transformations



$${}^A T_D = \begin{bmatrix} {}^A R_B {}^B R_C {}^C R_D & ({}^A R_B {}^B R_C {}^C P_D + {}^A R_B {}^B P_C + {}^A P_B) \\ \mathbf{O}_{1 \times 3} & 1 \end{bmatrix} \quad (50)$$

Exercise-2

Mobile manipulation with frame transformation.



Exercise-2

$${}^C\mathbf{T}_B = \begin{bmatrix} 0 & 0 & -1 & 3 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 1.2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^A\mathbf{T}_C = \begin{bmatrix} 0 & 0 & -1 & 4 \\ 0 & -1 & 0 & 0.5 \\ -1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (51)$$

$${}^R\mathbf{T}_E = \begin{bmatrix} 0 & -1/\sqrt{2} & -1/\sqrt{2} & 0.3 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} & -0.4 \\ 1 & 0 & 0 & 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^C\mathbf{T}_R = \begin{bmatrix} 0 & 0 & -1 & 2.5 \\ 0 & -1 & 0 & -1.5 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (52)$$

Find the position of the bottle in the robot end-effector coordinate system (i.e. find ${}^E\mathbf{T}_B$ first).

Solution

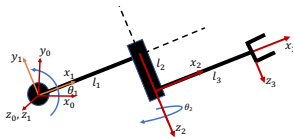
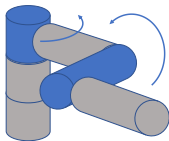
We can use the following relationship to find ${}^E\mathbf{T}_B$: Given ${}^C\mathbf{T}_B$, ${}^A\mathbf{T}_C$, ${}^R\mathbf{T}_E$, ${}^C\mathbf{T}_R$, We can write the following equation:

$$\begin{aligned} {}^C\mathbf{T}_R {}^R\mathbf{T}_E {}^E\mathbf{T}_B &= {}^C\mathbf{T}_B \\ {}^E\mathbf{T}_B &= ({}^C\mathbf{T}_R {}^R\mathbf{T}_E)^{-1} {}^C\mathbf{T}_B \end{aligned} \quad (53)$$

$${}^E\mathbf{T}_B = \begin{bmatrix} 0 & 0 & 1 & -0.75 \\ -0.7071 & 0.7071 & 0 & -1.8385 \\ -0.7071 & -0.7071 & 0 & 1.1314 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (54)$$

Then the position of the bottle in the end-effector coordinate system is $[-0.75, -1.8385, 1.1314]^T$

Example 1



The homogeneous transformation matrix of a 3-link (2 joints) robot manipulator is given by ${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3$

$${}^0T_3 = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ s\theta_2 & c\theta_2 & 0 & l_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (55)$$

Jacobian matrix from the position vector

Once we know the homogeneous transformation matrix for a robot manipulator, we can obtain the rotation matrix which gives the orientation of the last link relative to the base link, and the position of the end effector relative to the base coordinate system.

A matrix known as the Jacobian matrix can be derived from the position vector to solve the inverse kinematics problem.

Let the position vector of the end effector of a robot manipulator be given by $p = f(\theta)$, where θ is the joint angle vector. Then we can use the chain rule to obtain the speed vector of the end effector with respect to the joint speeds given by

$$\frac{dp}{dt} = \frac{\partial f(\theta)}{\partial \theta} \frac{d\theta}{dt} \quad (56)$$

Then the Jacobian matrix $J = \frac{\partial f(\theta)}{\partial \theta}$.

$$\dot{p} = J\dot{\theta} \quad (57)$$

Then, if J is a square matrix, we can obtain the joint angular speeds given an end-point speed vector using $\dot{\theta} = J^{-1}\dot{p}$. If this can be computed, a given desired end-effector position profile can first be differentiated to obtain the speed profile, and then the above relationship can be used to obtain the joint angular speed profile. Then the joint angle profile can be obtained by integrating the joint angular speed profile.

However, in most cases, the Jacobian matrix is not a square matrix due to more than 3-joints in the manipulator. In such cases, we use the Moore-Penrose pseudo-inverse of the Jacobian given by $J^+ = J^T(JJ^T)^{-1}$.

Example 1

We obtained that the position vector is given by

$${}^0\mathbf{P}_3 = \begin{bmatrix} l_1 c\theta_1 + l_2 s\theta_1 + l_3 c\theta_1 c\theta_2 \\ l_1 s\theta_1 - l_2 c\theta_1 + l_3 c\theta_2 s\theta_1 \\ l_3 s\theta_2 \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad (58)$$

Then the Jacobian is given by

$$J(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\partial p_x}{\partial \theta_1} & \frac{\partial p_x}{\partial \theta_2} \\ \frac{\partial p_y}{\partial \theta_1} & \frac{\partial p_y}{\partial \theta_2} \\ \frac{\partial p_z}{\partial \theta_1} & \frac{\partial p_z}{\partial \theta_2} \end{bmatrix} \quad (59)$$

Run JacobianMat.m to obtain the Jacobian matrix.

$$J(\boldsymbol{\theta}) = \begin{bmatrix} l_2 \cos(\theta_1) - l_1 \sin(\theta_1) - l_3 \cos(\theta_2) \sin(\theta_1), & -l_3 \cos(\theta_1) \sin(\theta_2) \\ l_1 \cos(\theta_1) + l_2 \sin(\theta_1) + l_3 \cos(\theta_1) \cos(\theta_2), & -l_3 \sin(\theta_1) \sin(\theta_2) \\ 0, & l_3 \cos(\theta_2) \end{bmatrix} \quad (60)$$

$$\dot{\boldsymbol{\theta}} = J^+(\boldsymbol{\theta})\dot{\boldsymbol{p}} \quad (61)$$

- 1 $J(\boldsymbol{\theta})$ has elements that are functions of $\boldsymbol{\theta}$.
- 2 Then, the map from $\dot{\boldsymbol{p}}$ to $\dot{\boldsymbol{\theta}}$ through the inverse jacobian matrix $J^+(\boldsymbol{\theta})$ depends on the joint angle vector $\boldsymbol{\theta}$ at any given time.

Landscape of elements in the pseudoinverse of J

Run `InvJacobianMat.m` to obtain the Jacobian matrix.

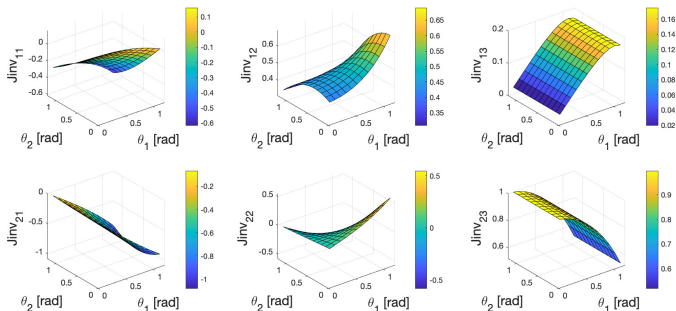
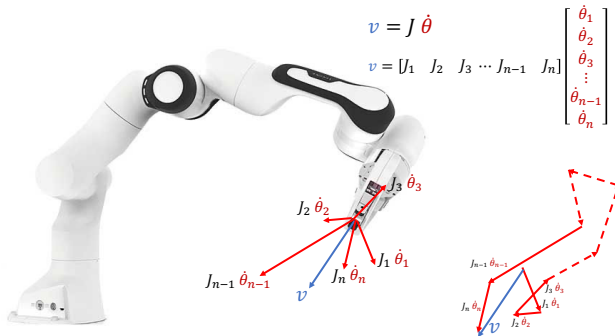


Figure: Elements of the $J^+(\theta)$ matrix for $\theta_1, \theta_2 \in [0, 1.2]$ [rad].

Some insights into the Jacobian

Each column of the Jacobian matrix is scaled by the corresponding joint angular velocity that results in the end effector velocity.

$$\begin{aligned}\dot{\mathbf{p}} &= \mathbf{J}\dot{\boldsymbol{\theta}} \\ \mathbf{v} &= \mathbf{J}\dot{\boldsymbol{\theta}}\end{aligned}\tag{62}$$



Mapping external forces to joint torques

In the case of a robotic manipulator, the work done by the torques around joints for a small joint movement is equal to the work done by the force vectors at contact points on the displacements of those points.

Virtual work done by the torque vector τ on a joint vector θ = virtual work done by the force vector f on a displacement vector p .

$$\begin{aligned}\tau^T \delta \theta &= f^T \delta p \\ \lim_{\delta t \rightarrow 0} \tau^T \frac{\delta \theta}{\delta t} &= \lim_{\delta t \rightarrow 0} f^T \frac{\delta p}{\delta t} \\ \tau^T \dot{\theta} &= f^T \dot{p} \\ \tau^T \dot{\theta} &= f^T J \dot{\theta} \\ \tau^T &= f^T J \\ \tau &= J^T f\end{aligned}\tag{63}$$

Example 1

For unit link lengths, use the Pseudoinverse of the Jacobian matrix to obtain the joint angles and angular velocities for unit velocity vectors at the end point, for two initial configurations:

$$\theta_1 = \pi/6, \theta_2 = \pi/6 \text{ and } \theta_1 = 0, \theta_2 = 0.$$

Hence make your own observations about how the joint angle configuration affects the joint velocity vector for a given end-effector velocity vector.